Since the equations (6.20) and (6.21) represent the same equations the coefficients of like terms are proportional.

$$\frac{\cos\alpha}{a}=\frac{\sin\alpha}{b}=\frac{p}{-c}\Rightarrow\frac{\cos\alpha}{a}=\frac{\sin\alpha}{b}=\frac{p}{-c}=\frac{\sqrt{\cos^2\alpha+\sin^2\alpha}}{\pm\sqrt{a^2+b^2}}$$

$$\cos\alpha=\frac{\pm a}{\sqrt{a^2+b^2}},\quad \sin\alpha=\frac{\pm b}{\sqrt{a^2+b^2}}\ \text{ and }\ p=\frac{\mp c}{\sqrt{a^2+b^2}}$$

Normal equation of CD is $\quad x\cos\alpha+y\sin\alpha=p'$ (6.22)

Since it passes through $P(x_1,y_1)$, (6.22) $\Rightarrow\ p'=x_1\cos\alpha+y_1\sin\alpha$

$$\begin{aligned}\text{Required distance} &= |PM|=|QR|=|OQ-OR| = |p'-p|\\ &= |(x_1\cos\alpha+y_1\sin\alpha)-p|\\ &= \left|\pm\frac{ax_1}{\sqrt{a^2+b^2}}\pm\frac{by_1}{\sqrt{a^2+b^2}}\pm\frac{c}{\sqrt{a^2+b^2}}\right|=\left|\pm\frac{ax_1+by_1+c}{\sqrt{a^2+b^2}}\right|\end{aligned}$$

$$\text{Required distance}=\left|\frac{ax_1+by_1+c}{\sqrt{a^2+b^2}}\right|$$

(iii) The distance between two parallel lines $a_1x+b_1y+c_1=0$ and $a_1x+b_1y+c_2=0$ is

$D=\dfrac{|c_2-c_1|}{\sqrt{a^2+b^2}}$ (It can be proved using above result by taking point (x_1,y_1) as the origin)

- The coordinates of the nearest point Q(foot of the perpendicular) on the line $ax+by+c=0$ from the point $P(x_1,y_1)$ can be found from

$$\frac{x-x_1}{a}=\frac{y-y_1}{b}=-\frac{(ax_1+by_1+c)}{a^2+b^2}\ \text{(using parametric form)}\quad (6.23)$$

- The coordinates of the image of the point $\bar{P}(x_1,y_1)$ with respect to the line $ax+by+c=0$ are given by

$$\frac{x-x_1}{a}=\frac{y-y_1}{b}=-\frac{2(ax_1+by_1+c)}{a^2+b^2}\quad (6.24)$$

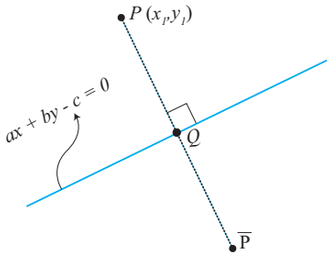


Figure 6.38

Example 6.22 Find the equations of a parallel line and a perpendicular line passing through the point (1, 2) to the line $3x+4y=7$.

Solution:

Parallel line to $3x+4y=7$ is of the form $3x+4y=3x_1+4y_1$ Let (x_1, y_1) be (1, 2)

$$\begin{aligned}\Rightarrow 3x+4y &= 3(1)+4(2)\\ 3x+4y &= 11\end{aligned}$$

Perpendicular line to $3x+4y=7$ is of the form

$$\begin{aligned}4x-3y &= 4x_1-3y_1\\ \text{Here } (x_1,y_1) &= (1,2)\\ \Rightarrow 4x-3y &= 4(1)-3(2)\\ 4x-3y &= -2\end{aligned}$$